Dataframe Joins

orderDF

order_id	prod_id	unit_price	qty
01	02	350	1
01	04	580	1
01	07	320	2
02	03	450	1
02	06	220	1
03	01	195	1
04	09	270	3
04	08	410	2
05	02	350	1

productDF

prod_id	name	list_price	qty
01	Scroll Mouse	250	20
02	Optical Mouse	350	20
03	Wireless Mouse	450	50
04	Wireless Keyboard	580	50
05	Standard Keyboard	360	10
06	16 GB Flash Storage	240	100
07	32 GB Flash Storage	320	50
08	64 GB Flash Storage	430	25

1. Join Condition/Expression : `join_expr = order_df.prod_id == product_df.prod_id`
2. Join Type : `joinType = "inner"`

```
order_df.join(product_renamed_df, join_expr, "inner")
```

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```python
join_expr = order_df.prod_id == product_df.prod_id

order_df.join(product_df, join_expr, "inner") \
    .show()
```

Run: SparkJoinDemo

```
+--------+-------+----------+---+-------+-------------------+----------+---+
|order_id|prod_id|unit_price|qty|prod_id|          prod_name|list_price|qty|
+--------+-------+----------+---+-------+-------------------+----------+---+
|      01|     07|       320|  2|     07|32 GB Flash Storage|       320| 50|
|      03|     01|       195|  1|     01|       Scroll Mouse|       250| 20|
|      04|     08|       410|  2|     08|64 GB Flash Storage|       430| 25|
|      02|     03|       450|  1|     03|     Wireless Mouse|       450| 50|
|      01|     02|       350|  1|     02|      Optical Mouse|       350| 20|
|      05|     02|       350|  1|     02|      Optical Mouse|       350| 20|
|      02|     06|       220|  1|     06|16 GB Flash Storage|       240|100|
|      01|     04|       580|  1|     04|  Wireless Keyboard|       580| 50|
+--------+-------+----------+---+-------+-------------------+----------+---+

Process finished with exit code 0
```

```python
join_expr = order_df.prod_id == product_df.prod_id

order_df.join(product_df, join_expr, "inner") \
    .select("order_id", "prod_name", "unit_price", "qty") \
    .show()
```

Run: SparkJoinDemo

```
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Traceback (most recent call last):
  File "C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/17-SparkJoinDemo/SparkJoinDemo.py", line 43, in <module>
    .select("order_id", "prod_name", "unit_price", "qty") \
  File "C:\Users\prashant\anaconda3\envs\PySpark3\lib\site-packages\pyspark\sql\dataframe.py", line 1421, in select
    jdf = self._jdf.select(self._jcols(*cols))
  File "C:\Users\prashant\anaconda3\envs\PySpark3\lib\site-packages\py4j\java_gateway.py", line 1305, in __call__
    answer, self.gateway_client, self.target_id, self.name)
  File "C:\Users\prashant\anaconda3\envs\PySpark3\lib\site-packages\pyspark\sql\utils.py", line 137, in deco
    raise_from(converted)
  File "<string>", line 3, in raise_from
pyspark.sql.utils.AnalysisException: <exception str() failed>

Process finished with exit code 1
```

Dataframe Joins

orderDF

order_id	prod_id	unit_price	qty
01	02	350	1
01	04	580	1
01	07	320	2
02	03	450	1
02	06	220	1
03	01	195	1
04	09	270	3
04	08	410	2
05	02	350	1

productDF

prod_id	name	list_price	qty
01	Scroll Mouse	250	20
02	Optical Mouse	350	20
03	Wireless Mouse	450	50
04	Wireless Keyboard	580	50
05	Standard Keyboard	360	10
06	16 GB Flash Storage	240	100
07	32 GB Flash Storage	320	50
08	64 GB Flash Storage	430	25

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File Edit View Navigate Code Refactor Run Tools VCS Window Help 17-SparkJoinDemo - SparkJoinDemo.py - PyCharm

17-SparkJoinDemo > SparkJoinDemo.py

SparkJoinDemo.py

```python
# product_df.show()
# order_df.show()

join_expr = order_df.prod_id == product_df.prod_id

product_renamed_df = product_df.withColumnRenamed("qty", "reorder_qty")

order_df.join(product_renamed_df, join_expr, "inner") \
    .drop(product_renamed_df.prod_id) \
    .select("order_id", "prod_id", "prod_name", "unit_price", "qty") \
    .show()
```

Run: SparkJoinDemo

```
+--------+-------+-------------------+----------+---+
|order_id|prod_id|          prod_name|unit_price|qty|
+--------+-------+-------------------+----------+---+
|      01|     07|32 GB Flash Storage|       320|  2|
|      03|     01|       Scroll Mouse|       195|  1|
|      04|     08|64 GB Flash Storage|       410|  2|
|      02|     03|     Wireless Mouse|       450|  1|
|      01|     02|      Optical Mouse|       350|  1|
|      05|     02|      Optical Mouse|       350|  1|
|      02|     06|16 GB Flash Storage|       220|  1|
|      01|     04|  Wireless Keyboard|       580|  1|
+--------+-------+-------------------+----------+---+


Process finished with exit code 0
```

9: Git 6: TODO 4: Run Terminal Python Console

Event Log

48:1 CRLF UTF-8 4 spaces Python 3.7 (PySpark3) master

Dataframe Joins

orderDF

order_id	prod_id	unit_price	qty
01	02	350	1
01	04	580	1
01	07	320	2
02	03	450	1
02	06	220	1
03	01	195	1
04	09	270	3
04	08	410	2
05	02	350	1

productDF

prod_id	name	list_price	qty
01	Scroll Mouse	250	20
02	Optical Mouse	350	20
03	Wireless Mouse	450	50
04	Wireless Keyboard	580	50
05	Standard Keyboard	360	10
06	16 GB Flash Storage	240	100
07	32 GB Flash Storage	320	50
08	64 GB Flash Storage	430	25

order_id	prod_id	prod_name	unit_price	list_price	qty
01	02	Optical Mouse	350	350	1
01	04	Wireless Keyboard	580	580	1
01	07	32 GB Flash Storage	320	320	2
02	03	Wireless Mouse	450	450	1
02	06	16 GB Flash Storage	220	240	1
03	01	Scroll Mouse	195	250	1
04	08	64 GB Flash Storage	410	430	2
05	02	Optical Mouse	350	350	1

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```python
product_renamed_df = product_df.withColumnRenamed("qty", "reorder_qty")

order_df.join(product_renamed_df, join_expr, "outer") \
    .drop(product_renamed_df.prod_id) \
    .select("*") \
    .sort("order_id") \
    .show()
```

if __name__ == "__main__"

Run: OuterJoinDemo

```
+--------+-------+----------+----+-------------------+----------+-----------+
|order_id|prod_id|unit_price| qty|          prod_name|list_price|reorder_qty|
+--------+-------+----------+----+-------------------+----------+-----------+
|    null|   null|      null|null|  Standard Keyboard|       360|         10|
|      01|     04|       580|   1|  Wireless Keyboard|       580|         50|
|      01|     07|       320|   2|32 GB Flash Storage|       320|         50|
|      01|     02|       350|   1|       Optical Mouse|       350|         20|
|      02|     03|       450|   1|      Wireless Mouse|       450|         50|
|      02|     06|       220|   1|16 GB Flash Storage|       240|        100|
|      03|     01|       195|   1|        Scroll Mouse|       250|         20|
|      04|     09|       270|   3|               null|      null|       null|
|      04|     08|       410|   2|64 GB Flash Storage|       430|         25|
|      05|     02|       350|   1|       Optical Mouse|       350|         20|
+--------+-------+----------+----+-------------------+----------+-----------+
```

Dataframe Joins

orderDF

order_id	prod_id	unit_price	qty
01	02	350	1
01	04	580	1
01	07	320	2
02	03	450	1
02	06	220	1
03	01	195	1
04	09	270	3
04	08	410	2
05	02	350	1

productDF

prod_id	name	list_price	qty
01	Scroll Mouse	250	20
02	Optical Mouse	350	20
03	Wireless Mouse	450	50
04	Wireless Keyboard	580	50
05	Standard Keyboard	360	10
06	16 GB Flash Storage	240	100
07	32 GB Flash Storage	320	50
08	64 GB Flash Storage	430	25

order_id	prod_id	unit_price	qty	prod_name	list_price	reorder_qty
null	null	null	null	Standard Keyboard	360	10
01	04	580	1	Wireless Keyboard	580	50
01	02	350	1	Optical Mouse	350	20
01	07	320	2	32 GB Flash Storage	320	50
02	06	220	1	16 GB Flash Storage	240	100
02	03	450	1	Wireless Mouse	450	50
03	01	195	1	Scroll Mouse	250	20
04	08	410	2	64 GB Flash Storage	430	25
04	09	270	3	null	null	null
05	02	350	1	Optical Mouse	350	20

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Dataframe Joins

orderDF

order_id	prod_id	unit_price	qty
01	02	350	1
01	04	580	1
01	07	320	2
02	03	450	1
02	06	220	1
03	01	195	1
04	09	270	3
04	08	410	2
05	02	350	1

1. Outer Join – Full Outer
2. Left Join – Left Outer
3. Right Join – Right Outer

productDF

prod_id	name	list_price	qty
01	Scroll Mouse	250	20
02	Optical Mouse	350	20
03	Wireless Mouse	450	50
04	Wireless Keyboard	580	50
05	Standard Keyboard	360	10
06	16 GB Flash Storage	240	100
07	32 GB Flash Storage	320	50
08	64 GB Flash Storage	430	25

order_id	prod_id	unit_price	qty	prod_name	list_price	reorder_qty
null	null	null	null	Standard Keyboard	360	10
01	04	580	1	Wireless Keyboard	580	50
01	02	350	1	Optical Mouse	350	20
01	07	320	2	32 GB Flash Storage	320	50
02	06	220	1	16 GB Flash Storage	240	100
02	03	450	1	Wireless Mouse	450	50
03	01	195	1	Scroll Mouse	250	20
04	08	410	2	64 GB Flash Storage	430	25
04	09	270	3	null	null	null
05	02	350	1	Optical Mouse	350	20

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```python
    product_renamed_df = product_df.withColumnRenamed("qty", "reorder_qty")

    order_df.join(product_renamed_df, join_expr, "left") \
        .drop(product_renamed_df.prod_id) \
        .select("order_id", "prod_id", "prod_name", "unit_price", "list_price", "qty") \
        .sort("order_id") \
        .show()
```

if __name__ == "__main__"

Run: OuterJoinDemo

```
C:\Users\prashant\anaconda3\envs\PySpark3\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/18-O
20/07/09 18:45:42 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
+--------+-------+-------------------+----------+----------+---+
|order_id|prod_id|          prod_name|unit_price|list_price|qty|
+--------+-------+-------------------+----------+----------+---+
|      01|     07|32 GB Flash Storage|       320|       320|  2|
|      01|     02|      Optical Mouse|       350|       350|  1|
|      01|     04|  Wireless Keyboard|       580|       580|  1|
|      02|     03|     Wireless Mouse|       450|       450|  1|
|      02|     06|16 GB Flash Storage|       220|       240|  1|
|      03|     01|       Scroll Mouse|       195|       250|  1|
|      04|     08|64 GB Flash Storage|       410|       430|  2|
|      04|     09|               null|       270|      null|  3|
|      05|     02|      Optical Mouse|       350|       350|  1|
+--------+-------+-------------------+----------+----------+---+
```

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18-OuterJoinDemo | OuterJoinDemo.py

OuterJoinDemo.py

```python

    join_expr = order_df.prod_id == product_df.prod_id

    product_renamed_df = product_df.withColumnRenamed("qty", "reorder_qty")

    order_df.join(product_renamed_df, join_expr, "left") \
        .drop(product_renamed_df.prod_id) \
        .select("order_id", "prod_id", "prod_name", "unit_price", "list_price", "qty") \
        .withColumn("prod_name", expr("coalesce(prod_name, prod_id)")) \
        .withColumn("list_price", expr("coalesce(list_price, unit_price)")) \
        .sort("order_id") \
        .show()
```

if __name__ == "__main__"

Run: OuterJoinDemo

```
C:\Users\prashant\anaconda3\envs\PySpark3\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/18-OuterJoinDemo/OuterJoinDemo.py
20/07/09 18:51:28 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
+--------+-------+-------------------+----------+----------+---+
|order_id|prod_id|          prod_name|unit_price|list_price|qty|
+--------+-------+-------------------+----------+----------+---+
|      01|     02|      Optical Mouse|       350|       350|  1|
|      01|     07|32 GB Flash Storage|       320|       320|  2|
|      01|     04|  Wireless Keyboard|       580|       580|  1|
|      02|     06|16 GB Flash Storage|       220|       240|  1|
|      02|     03|     Wireless Mouse|       450|       450|  1|
|      03|     01|       Scroll Mouse|       195|       250|  1|
|      04|     08|64 GB Flash Storage|       410|       430|  2|
|      04|     09|                 09|       270|       270|  3|
|      05|     02|      Optical Mouse|       350|       350|  1|
+--------+-------+-------------------+----------+----------+---+
```

9: Git 6: TODO 4: Run Terminal Python Console Event Log

50:16 CRLF UTF-8 4 spaces Python 3.7 (PySpark3) master

Join Internals

1. Shuffle Join
2. Broadcast Join

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Join Internals

Id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
00	1/1/2000	DL	1451	BOS	Boston, MA	ATL	Atlanta, GA
01	1/1/2000	DL	1479	BOS	Boston, MA	ATL	Atlanta, GA
02	1/1/2000	DL	1857	BOS	Boston, MA	ATL	Atlanta, GA
03	1/1/2000	DL	1997	BOS	Boston, MA	ATL	Atlanta, GA
04	1/1/2000	DL	2065	BOS	Boston, MA	ATL	Atlanta, GA
05	1/1/2000	US	2619	BOS	Boston, MA	ATL	Atlanta, GA
06	1/1/2000	US	2621	BOS	Boston, MA	ATL	Atlanta, GA
07	1/1/2000	DL	346	BTR	Baton Rouge, LA	ATL	Atlanta, GA
08	1/1/2000	DL	412	BTR	Baton Rouge, LA	ATL	Atlanta, GA
09	1/1/2000	DL	299	BUF	Buffalo, NY	ATL	Atlanta, GA
10	1/1/2000	DL	495	BUF	Buffalo, NY	ATL	Atlanta, GA

Id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
11	1/1/2000	DL	677	BUF	Buffalo, NY	ATL	Atlanta, GA
12	1/1/2000	DL	251	BWI	Baltimore, MD	ATL	Atlanta, GA
13	1/1/2000	DL	1003	BWI	Baltimore, MD	ATL	Atlanta, GA
14	1/1/2000	DL	1501	BWI	Baltimore, MD	ATL	Atlanta, GA
15	1/1/2000	DL	1907	BWI	Baltimore, MD	ATL	Atlanta, GA
16	1/1/2000	DL	2063	BWI	Baltimore, MD	ATL	Atlanta, GA
17	1/1/2000	DL	2111	BWI	Baltimore, MD	ATL	Atlanta, GA
18	1/1/2000	US	2632	BWI	Baltimore, MD	ATL	Atlanta, GA
19	1/1/2000	US	2967	BWI	Baltimore, MD	ATL	Atlanta, GA
20	1/1/2000	DL	540	CAE	Columbia, SC	ATL	Atlanta, GA

Id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
21	1/1/2000	DL	544	CAE	Columbia, SC	ATL	Atlanta, GA
22	1/1/2000	DL	1289	CAE	Columbia, SC	ATL	Atlanta, GA
23	1/1/2000	DL	1518	CAE	Columbia, SC	ATL	Atlanta, GA
24	1/1/2000	DL	1519	CAE	Columbia, SC	ATL	Atlanta, GA
25	1/1/2000	DL	1564	CAE	Columbia, SC	ATL	Atlanta, GA
26	1/1/2000	DL	2030	CAE	Columbia, SC	ATL	Atlanta, GA
27	1/1/2000	DL	391	CHS	Charleston, SC	ATL	Atlanta, GA
28	1/1/2000	DL	423	CHS	Charleston, SC	ATL	Atlanta, GA
29	1/1/2000	DL	717	CHS	Charleston, SC	ATL	Atlanta, GA
30	1/1/2000	DL	771	CHS	Charleston, SC	ATL	Atlanta, GA

E1

E2

E3

Id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
00	1115	1113	1343	5	1400	1348	0	946
01	1315	1311	1536	7	1559	1543	0	946
02	1415	1414	1642	9	1721	1651	0	946
03	1715	1720	1955	10	2013	2005	0	946
04	2015	2010	2230	10	2300	2240	0	946
05	650	649	956	7	955	1003	0	946
06	1440	1446	1713	4	1738	1717	0	946
07	1740	1744	1957	9	2008	2006	0	449
08	1345	1345	1552	9	1622	1601	0	449
09	1245	1245	1443	5	1455	1448	0	712
10	2035	2035	2226	9	2241	2235	0	712

Id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
11	710	710	940	7	925	947	0	712
12	2040	2100	2235	7	2233	2242	0	576
13	1635	1838	2020	12	1832	2032	0	576
14	1430	1435	1623	12	1634	1635	0	576
15	530	530	716	4	723	720	0	576
16	1250				1449		1	576
17	1845	1855	2041	9	2046	2050	0	576
18	710	710			905		0	576
19	1700	1700	1845	6	1853	1851	0	576
20	955	952	1052	5	1104	1057	0	191

Id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
21	2130	2125	2208	10	2232	2218	0	191
22	1515	1514	1619	9	1633	1628	0	191
23	1735	1737	1844	7	1852	1851	0	191
24	720	720	811	10	818	821	0	191
25	1935	1931	2019	9	2047	2028	0	191
26	1340				1450		1	191
27	1005	1002	1100	7	1118	1107	0	259
28	600	556	654	17	707	711	0	259
29	1340	1340	1432	7	1454	1439	0	259
30	2135	2125	2219	5	2242	2224	0	259

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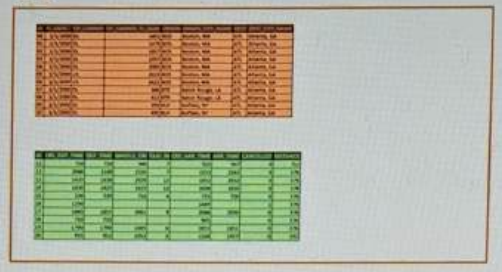
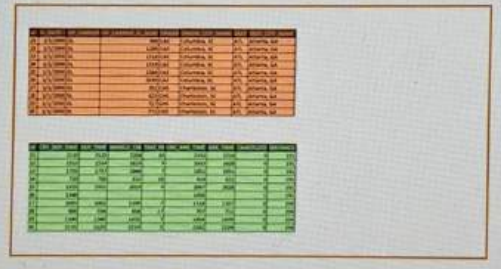
Join Internals

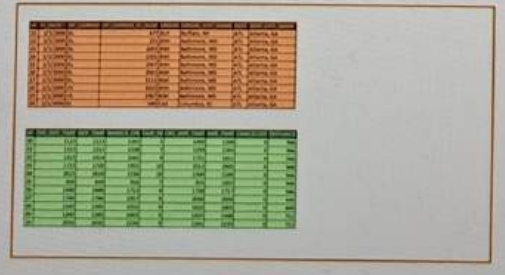
E1



E2



E3



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Join Internals

E1

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
00	1/1/2000	DL	1451	BOS	Boston, MA	ATL	Atlanta, GA
01	1/1/2000	DL	1479	BOS	Boston, MA	ATL	Atlanta, GA
02	1/1/2000	DL	1857	BOS	Boston, MA	ATL	Atlanta, GA
03	1/1/2000	DL	1997	BOS	Boston, MA	ATL	Atlanta, GA
04	1/1/2000	DL	2065	BOS	Boston, MA	ATL	Atlanta, GA
05	1/1/2000	US	2629	BOS	Boston, MA	ATL	Atlanta, GA
06	1/1/2000	US	2621	BOS	Boston, MA	ATL	Atlanta, GA
07	1/1/2000	DL	346	BTR	Baton Rouge, LA	ATL	Atlanta, GA
08	1/1/2000	DL	412	BTR	Baton Rouge, LA	ATL	Atlanta, GA
09	1/1/2000	DL	299	BUF	Buffalo, NY	ATL	Atlanta, GA
10	1/1/2000	DL	485	BUF	Buffalo, NY	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
11	720	716	940	7	925	947	0	712
12	2040	2100	2235	7	2233	2242	0	576
13	1635	1638	2020	12	1832	2032	0	576
14	1430	1435	1621	12	1634	1625	0	576
15	530	530	716	4	723	720	0	576
16	1250				1449		1	576
17	1845	1830	2041	9	2046	2050	0	576
18	710	710			905		0	576
19	1700	1700	1845	6	1853	1851	0	576
20	955	952	1052	5	1104	1057	0	191

E2

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
21	1/1/2000	DL	544	CAE	Columbia, SC	ATL	Atlanta, GA
22	1/1/2000	DL	1289	CAE	Columbia, SC	ATL	Atlanta, GA
23	1/1/2000	DL	1518	CAE	Columbia, SC	ATL	Atlanta, GA
24	1/1/2000	DL	1519	CAE	Columbia, SC	ATL	Atlanta, GA
25	1/1/2000	DL	1564	CAE	Columbia, SC	ATL	Atlanta, GA
26	1/1/2000	DL	2030	CAE	Columbia, SC	ATL	Atlanta, GA
27	1/1/2000	DL	391	CHS	Charleston, SC	ATL	Atlanta, GA
28	1/1/2000	DL	423	CHS	Charleston, SC	ATL	Atlanta, GA
29	1/1/2000	DL	717	CHS	Charleston, SC	ATL	Atlanta, GA
30	1/1/2000	DL	771	CHS	Charleston, SC	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
21	2130	2125	2208	10	2232	2218	0	191
22	1515	1514	1619	9	1633	1628	0	191
23	1735	1737	1844	7	1852	1851	0	191
24	720	720	811	10	818	821	0	191
25	1935	1931	2019	9	2047	2028	0	191
26	1340				1450		1	191
27	1005	1002	1100	7	1118	1107	0	259
28	600	556	654	17	707	711	0	259
29	1340	1340	1432	7	1454	1439	0	259
30	2135	2125	2219	5	2242	2224	0	259

E3

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
11	1/1/2000	DL	677	BUF	Buffalo, NY	ATL	Atlanta, GA
12	1/1/2000	DL	251	BWI	Baltimore, MD	ATL	Atlanta, GA
13	1/1/2000	DL	1003	BWI	Baltimore, MD	ATL	Atlanta, GA
14	1/1/2000	DL	1501	BWI	Baltimore, MD	ATL	Atlanta, GA
15	1/1/2000	DL	1907	BWI	Baltimore, MD	ATL	Atlanta, GA
16	1/1/2000	DL	2063	BWI	Baltimore, MD	ATL	Atlanta, GA
17	1/1/2000	DL	2111	BWI	Baltimore, MD	ATL	Atlanta, GA
18	1/1/2000	US	2632	BWI	Baltimore, MD	ATL	Atlanta, GA
19	1/1/2000	US	2967	BWI	Baltimore, MD	ATL	Atlanta, GA
20	1/1/2000	DL	540	CAE	Columbia, SC	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
00	1115	1113	1343	5	1400	1348	0	946
01	1315	1311	1536	7	1559	1543	0	946
02	1415	1414	1642	9	1721	1651	0	946
03	1715	1720	1955	10	2013	2005	0	946
04	2015	2010	2230	10	2300	2240	0	946
05	650	649	956	7	955	1003	0	946
06	1440	1446	1713	4	1738	1717	0	946
07	1740	1744	1957	9	2008	2006	0	449
08	1345	1345	1552	9	1622	1601	0	449
09	1245	1245	1443	5	1455	1448	0	712
10	2035	2035	2226	9	2241	2235	0	712

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Join Internals

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Join Internals

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
00	1/1/2000	DL	1451	BOS	Boston, MA	ATL	Atlanta, GA
01	1/1/2000	DL	1479	BOS	Boston, MA	ATL	Atlanta, GA
02	1/1/2000	DL	1857	BOS	Boston, MA	ATL	Atlanta, GA
03	1/1/2000	DL	1997	BOS	Boston, MA	ATL	Atlanta, GA
04	1/1/2000	DL	2065	BOS	Boston, MA	ATL	Atlanta, GA
05	1/1/2000	US	2829	BOS	Boston, MA	ATL	Atlanta, GA
06	1/1/2000	DL	2621	BOS	Boston, MA	ATL	Atlanta, GA
07	1/1/2000	DL	340	BTR	Baton Rouge, LA	ATL	Atlanta, GA
08	1/1/2000	DL	412	BTR	Baton Rouge, LA	ATL	Atlanta, GA
09	1/1/2000	DL	299	BUF	Buffalo, NY	ATL	Atlanta, GA
10	1/1/2000	DL	495	BUF	Buffalo, NY	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
00	1113	1113	1343	5	1400	1348	0	946
01	1315	1311	1536	7	1559	1543	0	946
02	1415	1414	1642	9	1721	1651	0	946
03	1715	1720	1955	10	2013	2005	0	946
04	2015	2010	2230	10	2300	2240	0	946
05	650	649	956	7	955	1003	0	946
06	1440	1446	1713	4	1738	1717	0	946
07	1740	1744	1957	9	2008	2006	0	449
08	1345	1345	1552	9	1622	1601	0	449
09	1245	1245	1443	5	1455	1448	0	712
10	2035	2035	2226	9	2241	2235	0	712

R1

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
21	1/1/2000	DL	544	CAE	Columbia, SC	ATL	Atlanta, GA
22	1/1/2000	DL	1285	CAE	Columbia, SC	ATL	Atlanta, GA
23	1/1/2000	DL	1536	CAE	Columbia, SC	ATL	Atlanta, GA
24	1/1/2000	DL	2539	CAE	Columbia, SC	ATL	Atlanta, GA
25	1/1/2000	DL	1564	CAE	Columbia, SC	ATL	Atlanta, GA
26	1/1/2000	DL	2030	CAE	Columbia, SC	ATL	Atlanta, GA
27	1/1/2000	DL	391	CHS	Charleston, SC	ATL	Atlanta, GA
28	1/1/2000	DL	423	CHS	Charleston, SC	ATL	Atlanta, GA
29	1/1/2000	DL	717	CHS	Charleston, SC	ATL	Atlanta, GA
30	1/1/2000	DL	772	CHS	Charleston, SC	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
21	2130	2125	2208	10	2232	2218	0	191
22	1515	1514	1619	9	1633	1628	0	191
23	1735	1737	1844	7	1852	1851	0	191
24	720	720	811	10	818	821	0	191
25	1935	1931	2019	9	2047	2028	0	191
26	1340				1450		1	191
27	1005	1002	1100	7	1118	1107	0	259
28	600	556	654	17	707	711	0	259
29	1340	1340	1432	7	1454	1439	0	259
30	2135	2125	2219	5	2242	2224	0	259

R2

id	FL_DATE	OP_CARRIER	OP_CARRIER_FL_NUM	ORIGIN	ORIGIN_CITY_NAME	DEST	DEST_CITY_NAME
11	1/1/2000	DL	677	BUF	Buffalo, NY	ATL	Atlanta, GA
12	1/1/2000	DL	251	BWI	Baltimore, MD	ATL	Atlanta, GA
13	1/1/2000	DL	1003	BWI	Baltimore, MD	ATL	Atlanta, GA
14	1/1/2000	DL	1501	BWI	Baltimore, MD	ATL	Atlanta, GA
15	1/1/2000	DL	1907	BWI	Baltimore, MD	ATL	Atlanta, GA
16	1/1/2000	DL	2063	BWI	Baltimore, MD	ATL	Atlanta, GA
17	1/1/2000	DL	2111	BWI	Baltimore, MD	ATL	Atlanta, GA
18	1/1/2000	US	2632	BWI	Baltimore, MD	ATL	Atlanta, GA
19	1/1/2000	US	2967	BWI	Baltimore, MD	ATL	Atlanta, GA
20	1/1/2000	DL	540	CAE	Columbia, SC	ATL	Atlanta, GA

id	CRS_DEP_TIME	DEP_TIME	WHEELS_ON	TAXI_IN	CRS_ARR_TIME	ARR_TIME	CANCELLED	DISTANCE
11	710	710	940	7	925	947	0	712
12	2040	2100	2235	7	2233	2242	0	576
13	1635	1838	2020	12	1832	2032	0	576
14	1430	1435	1623	12	1634	1635	0	576
15	530	530	716	4	723	720	0	576
16	1250				1449		1	576
17	1845	1855	2041	9	2046	2050	0	576
18	710	710			905		0	576
19	1700	1700	1845	6	1853	1851	0	576
20	955	952	1052	5	1104	1057	0	191

R3

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File Edit View Navigate Code Refactor Run Tools VCS Window Help 19-ShuffleJoinDemo - SuffleJoinDemo.py - PyCharm

19-ShuffleJoinDemo

Project

19-ShuffleJoinDemo C:\Users\prashant\Documents\GitHub\Spark-Programmi

- app-logs
- data
  - d1
    - part-00000-00af64b6-7ef5-4909-8f82-b8897114efaf-c000.json
    - part-00001-00af64b6-7ef5-4909-8f82-b8897114efaf-c000.json
    - part-00002-00af64b6-7ef5-4909-8f82-b8897114efaf-c000.json
  - d2
    - part-00000-ee8814fd-bbb7-4a16-912b-f74b04a6d4fa-c000.json
    - part-00001-ee8814fd-bbb7-4a16-912b-f74b04a6d4fa-c000.json
    - part-00002-ee8814fd-bbb7-4a16-912b-f74b04a6d4fa-c000.json
- lib
- log4j.properties
- SuffleJoinDemo.py
- External Libraries
- Scratches and Consoles

SuffleJoinDemo.py

```python
import ...

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .appName("Shuffle Join Demo") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4j(spark)

    flight_time_df1 = spark.read.json("data/d1/")
    flight_time_df2 = spark.read.json("data/d2/")
```

```python
logger = Log4j(spark)

flight_time_df1 = spark.read.json("data/d1/")
flight_time_df2 = spark.read.json("data/d2/")

spark.conf.set("spark.sql.shuffle.partitions", 3)

join_expr = flight_time_df1.id == flight_time_df2.id
join_df = flight_time_df1.join(flight_time_df2, join_expr, "inner")

join_df.foreach(lambda f: None)
input("press a key to stop...")
```

Shuffle Join Demo - Spark Jobs

localhost:4040/jobs/

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Shuffle Join Demo application UI

Spark Jobs (?)

User: prashant
Total Uptime: 32 s
Scheduling Mode: FIFO
Completed Jobs: 3

▸ Event Timeline

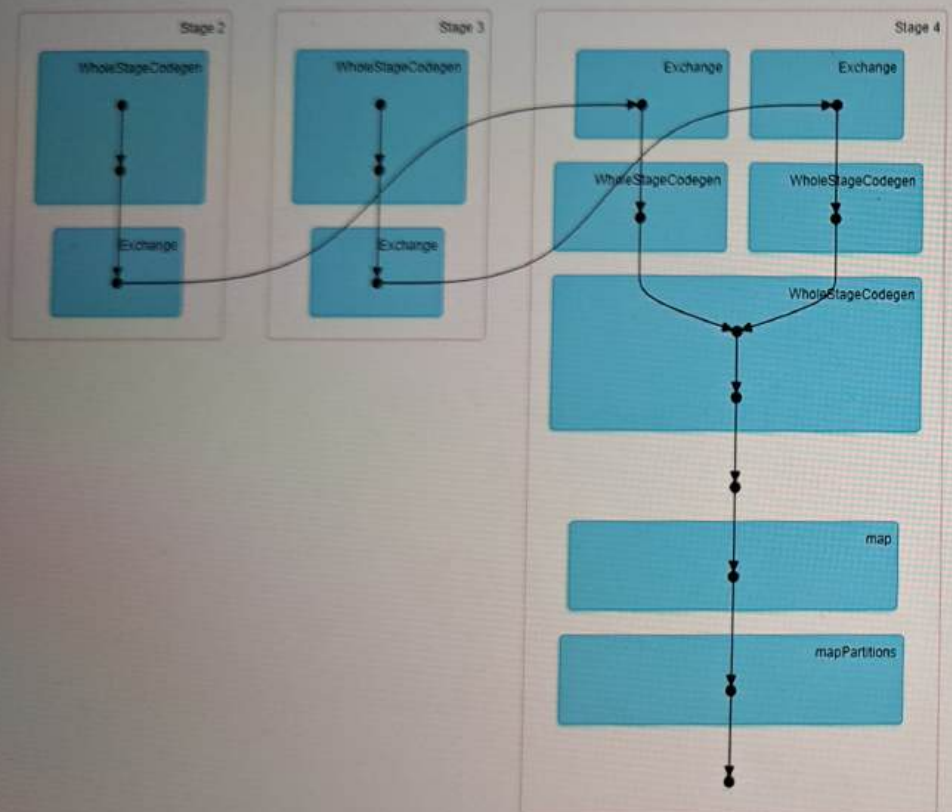
▾ Completed Jobs (3)

Job Id ▾	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	foreach at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/19-ShuffleJoinDemo/SuffleJoinDemo.py:22 foreach at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/19-ShuffleJoinDemo/SuffleJoinDemo.py:22	2020/07/10 19:42:46	7 s	3/3	9/9
1	json at NativeMethodAccessorImpl.java:0 json at NativeMethodAccessorImpl.java:0	2020/07/10 19:42:45	0.6 s	1/1	3/3
0	json at NativeMethodAccessorImpl.java:0 json at NativeMethodAccessorImpl.java:0	2020/07/10 19:42:44	1.0 s	1/1	3/3

Shuffle Join Demo - Details for

localhost:4040/jobs/job/?id=2

DAG Visualization



Completed Stages (3)

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
4	foreach at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/19-ShuffleJoinDemo/SuffleJoinDemo.py:22 +details	2020/07/10 19:42:48	5 s	3/3			24.4 MB	
3	javaToPython at NativeMethodAccessorImpl.java:0 +details	2020/07/10 19:42:46	0.6 s	3/3	64.0 MB			13.9 MB
2	javaToPython at NativeMethodAccessorImpl.java:0 +details	2020/07/10 19:42:46	1 s	3/3	79.4 MB			10.5 MB

Join Internals

1. Large to Large
2. Large to Small

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Join Internals

1. Don't code like a novice
2. Shuffle partitions and Parallelism
3. Key Distribution

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Join Internals

global_online_sales

order_id	city	product_code	store_code
S001	New York	ABX-512	abc.com
~~S002~~	~~London~~	~~GCY-213~~	~~abc.com~~

country_city

country	city
US	New York
~~UK~~	~~London~~

us_warehouse_events

order_id	city	warehouse_code	event_type	event_date
S001	New York	NY516	order placed	15-Jan-2020
S001	New York	NY516	dispatched	16-Jan-2020

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Join Internals

1. Don't code like a novice
2. Shuffle partitions and Parallelism
3. Key Distribution

   2.1 Shuffle Partitions
   2.2 Number of Executors

What is the maximum possible parallelism?
Executors = 500
Shuffle Partitions = 400
Unique Keys = 200

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Join Internals

```
sales
----------
order_id, product_id, units, price, ....
```

```
products
--------------------------
product_id, product_name, list_price, ....
```

200 Unique Products
Max Shuffle Partitions = 200
→ Max Parallel Tasks = 200

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Join Internals

1. Don't code like a novice
2. Shuffle partitions and Parallelism
3. Key Distribution

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Join Internals

Join

```
products
-------------------------
product_id, product_name, list_price, ....
```

```
sales
----------
order_id, product_id, units, price, ....
```

Shuffle

Fast Moving Product

Slow Moving Product

Very Slow Moving Product

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Join Internals

1. Huge Volumes – Filter/Aggregate
2. Parallelism – Shuffles/Executors/Keys
3. Shuffle Distribution – Key Skews

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Join Internals

1. Large to Large – Shuffle Join
2. Large to Small – Broadcast Join

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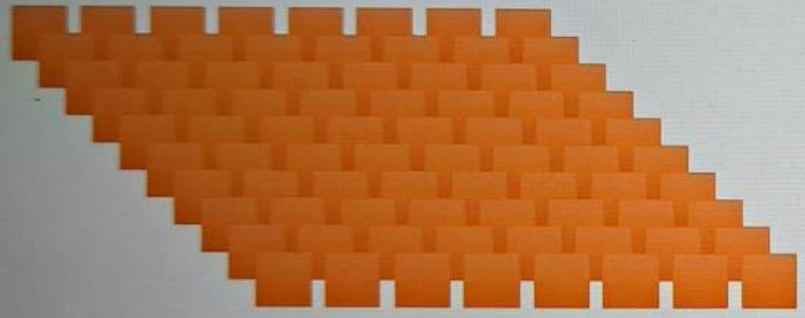
Join Internals

```
sales
----------
order_id, product_id, units, price, ....
```



```
products
--------------------------
product_id, product_name, list_price, ....
```

Shuffle

Exchange

Exchange

Exchange

Exchange

Exchange

Exchange

Exchange

Exchange

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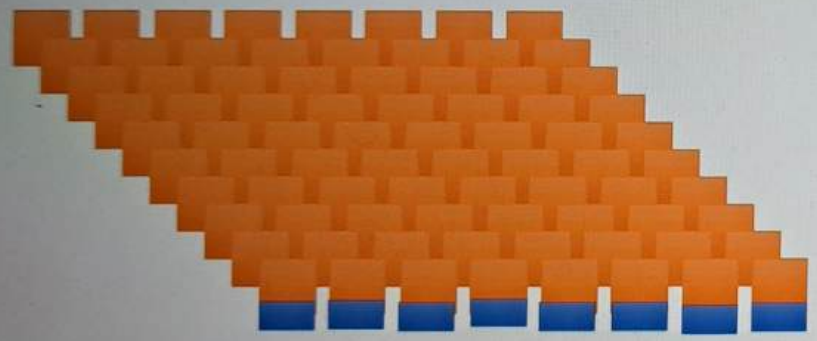
Join Internals

```
sales
----------
order_id, product_id, units, price, ....
```



```
products
--------------------------
product_id, product_name, list_price, ....
```

Spark-submit options

--driver-memory

--executor-memory

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How to broadcast join?

udemy

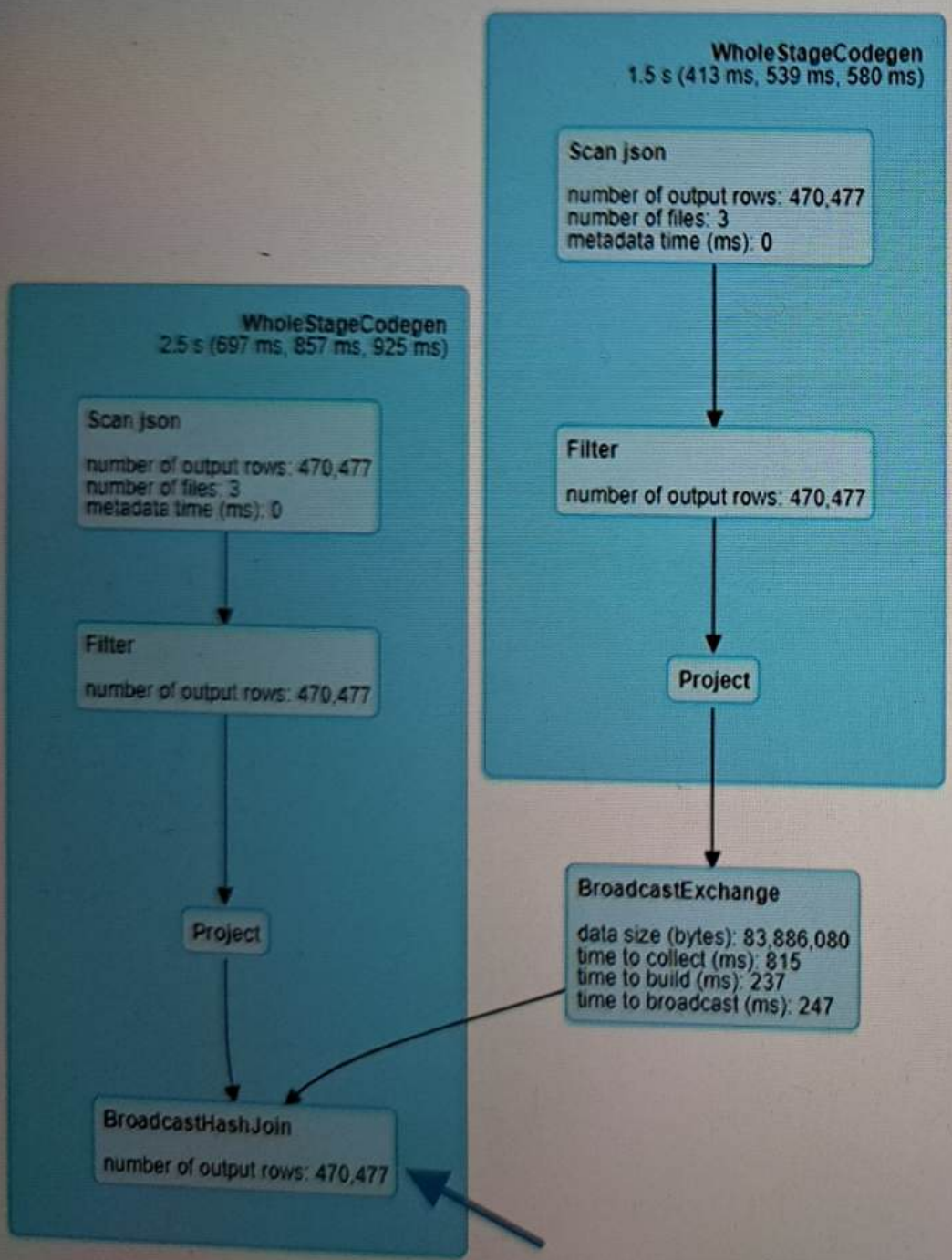
SuffleJoinDemo.py

```python
from pyspark.sql import SparkSession
from pyspark.sql.functions import broadcast
from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .appName("Shuffle Join Demo") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4j(spark)

    flight_time_df1 = spark.read.json("data/d1/")
    flight_time_df2 = spark.read.json("data/d2/")

    spark.conf.set("spark.sql.shuffle.partitions", 3)

    join_expr = flight_time_df1.id == flight_time_df2.id
    join_df = flight_time_df1.join(broadcast(flight_time_df2), join_expr, "inner")

    join_df.foreach(lambda f: None)
    input("press a key to stop...")
```

Spark 2.4.5 Jobs Stages Storage Environment Executors SQL

Details for Query 0

Submitted Time: 2020/07/13 12:12:42
Duration: 3 s
Succeeded Jobs: 2 3



Details

Project

- 19-ShuffleJoinDemo C:\Users\prashant\D
  - app-logs
  - data
    - d1
    - d2
  - lib
  - log4j.properties
  - SuffleJoinDemo.py
- External Libraries
- Scratches and Consoles

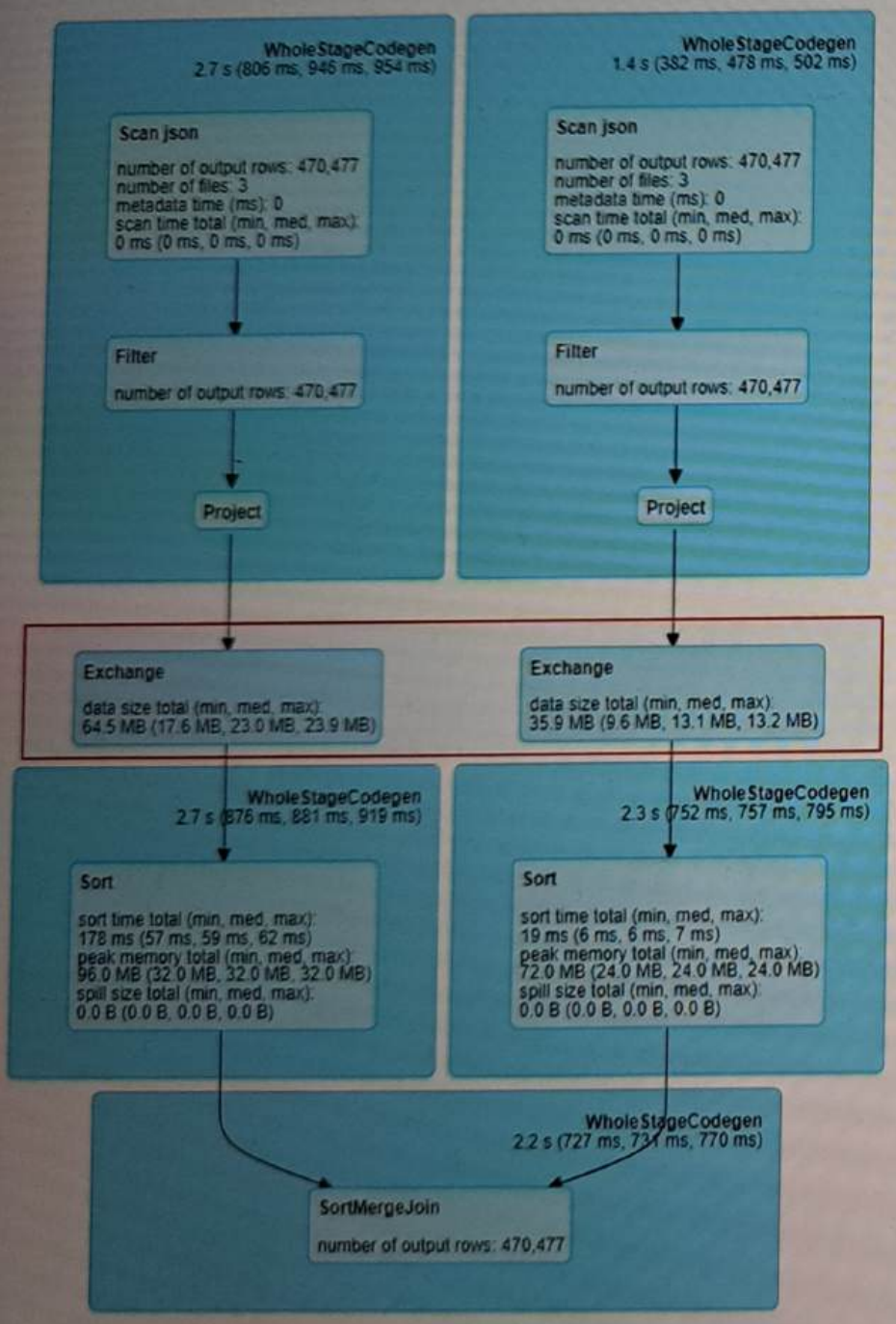
SuffleJoinDemo.py

```python
from pyspark.sql import SparkSession

from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .appName("Shuffle Join Demo") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4j(spark)

    flight_time_df1 = spark.read.json("data/d1/")
    flight_time_df2 = spark.read.json("data/d2/")

    spark.conf.set("spark.sql.shuffle.partitions", 3)

    join_expr = flight_time_df1.id == flight_time_df2.id
    join_df = flight_time_df1.join(flight_time_df2, join_expr, "inner")

    join_df.collect()
    input("press a key to stop...")
```

Shuffle Join Demo - Details for C

localhost:4040/SQL/execution/?id=0

Succeeded Jobs: 2



Details

File Edit View Navigate Code Refactor Run Tools VCS Window Help 20-BucketJoinDemo - BucketJoinDemo.py - PyCharm

20-BucketJoinDemo > BucketJoinDemo.py

Project

20-BucketJoinDemo C:\Users\prashant\Documents\GitHub\Sp
- data
  - d1
  - d2
- lib
- BucketJoinDemo.py
- log4j.properties
- External Libraries
- Scratches and Consoles

BucketJoinDemo.py

```python
from pyspark.sql import SparkSession

from lib.logger import Log4j

if __name__ == "__main__":
    Run 'BucketJoinDemo'  Ctrl+Shift+F10
    Debug 'BucketJoinDemo'
    Edit 'BucketJoinDemo'...
        .appName("Bucket Join Demo") \
        .master("local[3]") \
        .enableHiveSupport() \
        .getOrCreate()

    logger = Log4j(spark)
    df1 = spark.read.json("data/d1/")
    df2 = spark.read.json("data/d2/")
    # df1.show()
    # df2.show()

    spark.sql("CREATE DATABASE IF NOT EXISTS MY_DB")
    spark.sql("USE MY_DB")

    df1.coalesce(1).write \
        .bucketBy(3, "id") \
        .mode("overwrite") \
        .saveAsTable("MY_DB.flight_data1")

    df2.coalesce(1).write \
        .bucketBy(3, "id") \
        .mode("overwrite") \
        .saveAsTable("MY_DB.flight_data2")
```

Git TODO Run Terminal Python Console

Event Log

31:1 CRLF UTF-8 4 spaces Python 3.7 (pyspark45) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 20-BucketJoinDemo - BucketJoinDemo.py - PyCharm

20-BucketJoinDemo / spark-warehouse

Project

- _SUCCESS.crc
- .part-00000-23e88903-9488-4585-8930-c9b34111b029_00000.c000.snappy.parquet.crc
- .part-00000-23e88903-9488-4585-8930-c9b34111b029_00001.c000.snappy.parquet.crc
- .part-00000-23e88903-9488-4585-8930-c9b34111b029_00002.c000.snappy.parquet.crc
- _SUCCESS
- part-00000-23e88903-9488-4585-8930-c9b34111b029_00000.c000.snappy.parquet
- part-00000-23e88903-9488-4585-8930-c9b34111b029_00001.c000.snappy.parquet
- part-00000-23e88903-9488-4585-8930-c9b34111b029_00002.c000.snappy.parquet
- flight_data2
  - _SUCCESS.crc
  - .part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00000.c000.snappy.parquet.crc
  - .part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00001.c000.snappy.parquet.crc
  - .part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00002.c000.snappy.parquet.crc
  - _SUCCESS
  - part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00000.c000.snappy.parquet
  - part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00001.c000.snappy.parquet
  - part-00000-7a722be5-e270-4d81-934c-78a5ea3bfc3c_00002.c000.snappy.parquet
- BucketJoinDemo.py
- derby.log
- log4j.properties

BucketJoinDemo.py

```
16      # df1.show()
17      # df2.show()
18
19      spark.sql("CREATE DATABASE IF NOT EXISTS MY_DB")
20      spark.sql("USE MY_DB")
21
22      df1.coalesce(1).write \
23          .bucketBy(3, "id") \
24          .mode("overwrite") \
25          .saveAsTable("MY_DB.flight_data1")
26
27      df2.coalesce(1).write \
28          .bucketBy(3, "id") \
29          .mode("overwrite") \
30          .saveAsTable("MY_DB.flight_data2")
31
32
33
```

```python
logger = Log4j(spark)
df1 = spark.read.json("data/d1/")
df2 = spark.read.json("data/d2/")
...
'''...'''

df3 = spark.read.table("MY_DB.flight_data1")
df4 = spark.read.table("MY_DB.flight_data2")

spark.conf.set("spark.sql.autoBroadcastJoinThreshold", -1)
join_expr = df3.id == df4.id
join_df = df3.join(df4, join_expr, "inner")

join_df.collect()
input("press a key to stop...")
```

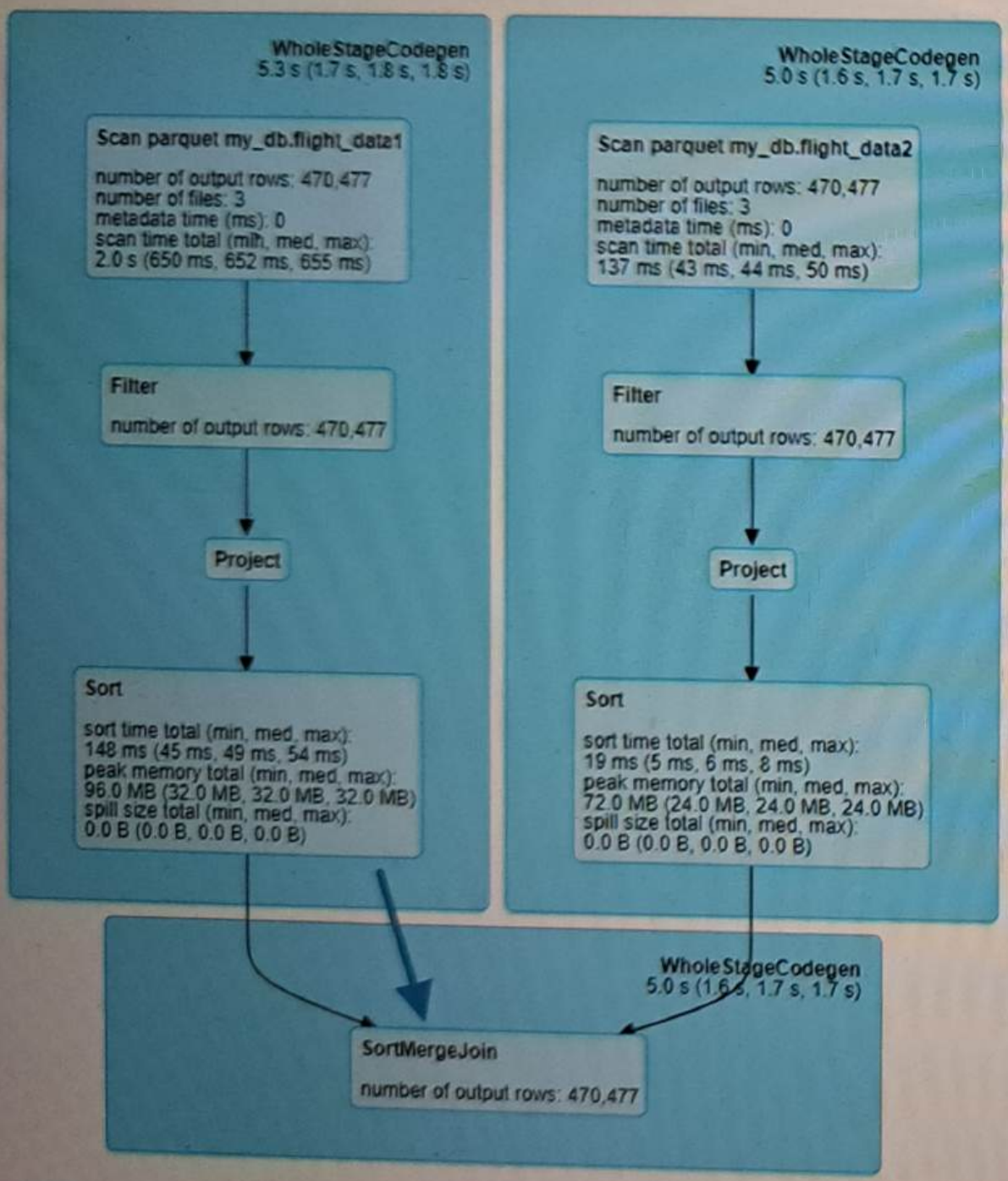
Bucket Join Demo - Details for Q

localhost:4040/SQL/execution/?id=0

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL

Details for Query 0

Submitted Time: 2020/07/14 18:31:08
Duration: 3 s
Succeeded Jobs: 2



Details